

Scala For DS

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Class : SYIT	Assignment Date : 11/08/2026

Practical no.14

14) Perform basic time series analysis in Scala. Generate synthetic time series data (e.g., daily sales over a month).

INPUT:

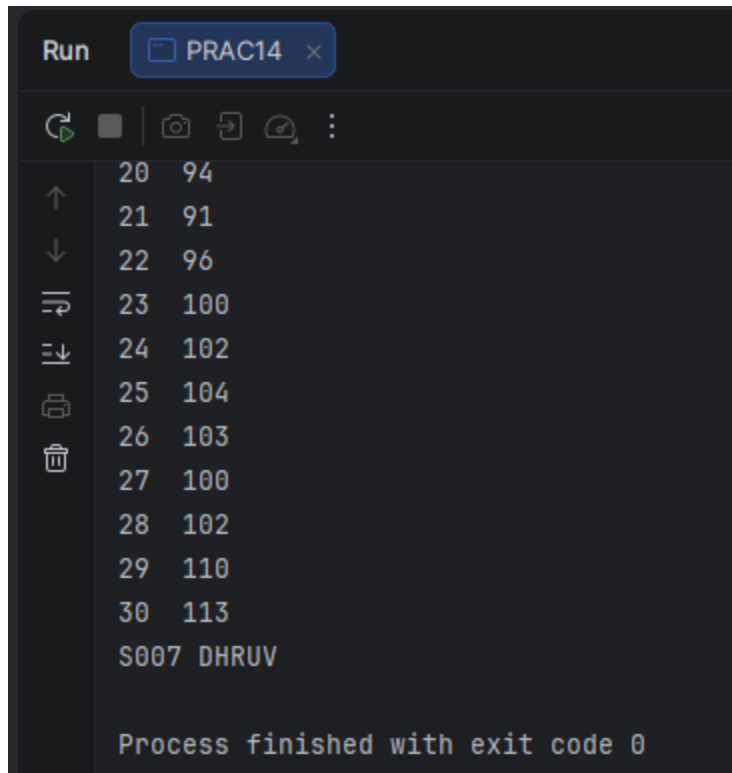
```
import scala.util.Random

object PRAC14 extends App {

  println("Day\tsales")

  for (day <- 1 to 30) {
    val sales = 50 + day * 2 + Random.nextInt(10) - 5
    println(s"$day\t$sales")
  }
  println("S007 DHRUV")
}
```

OUTPUT:



```
Run PRAC14 x
20 94
21 91
22 96
23 100
24 102
25 104
26 103
27 100
28 102
29 110
30 113
S007 DHRUV
Process finished with exit code 0
```

Practical no.15

15) Create polynomial features from a dataset. Given a list of numbers (e.g., [1, 2, 3]), generate polynomial features up to degree 3.

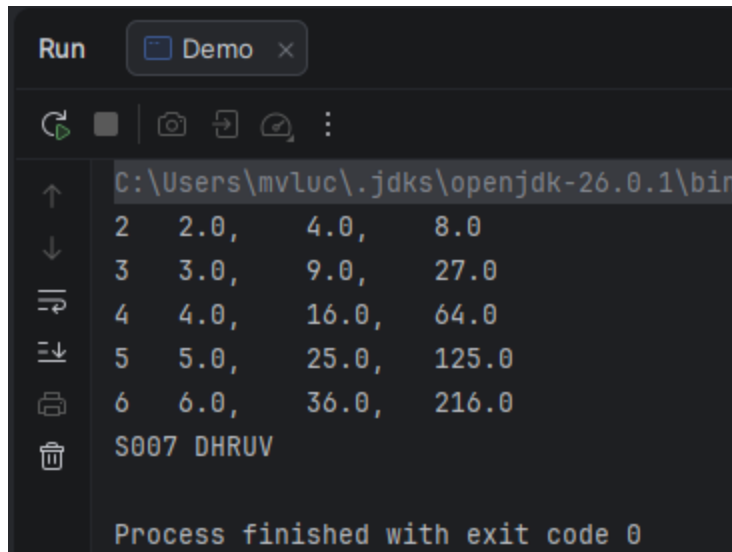
INPUT:

```
object PRAC14 extends App {
  val data = List(2, 3, 4, 5, 6)

  for (x <- data) {
    val p1 = math.pow(x, 1)
    val p2 = math.pow(x, 2)
    val p3 = math.pow(x, 3)

    println(s"$x\t$p1, \t$p2, \t$p3")
  }
  println("S007 DHRUV")
}
```

OUTPUT:



The screenshot shows a terminal window with a title bar 'Run Demo'. The terminal output is as follows:

```
C:\Users\mvluc\.jdk\openjdk-26.0.1\bin
2 2.0, 4.0, 8.0
3 3.0, 9.0, 27.0
4 4.0, 16.0, 64.0
5 5.0, 25.0, 125.0
6 6.0, 36.0, 216.0
S007 DHRUV
Process finished with exit code 0
```

Practical no.16

16) Create a scatter plot of random data using Breeze-viz. Label the axes and customize the color of points.

INPUT:

```
libraryDependencies += "org.scalanlp" %% "breeze" % "2.1.0"
libraryDependencies += "org.scalanlp" %% "breeze-viz" % "2.1.0"
```

```

import breeze.plot._
import scala.util.Random

object PRAC14 extends App {
  val x = List.fill(10)(Random.nextDouble() * 10)
  val y = List.fill(10)(Random.nextDouble() * 20)

  println(x)
  println(y)

  val f = Figure()
  val p = f.subplot(0)

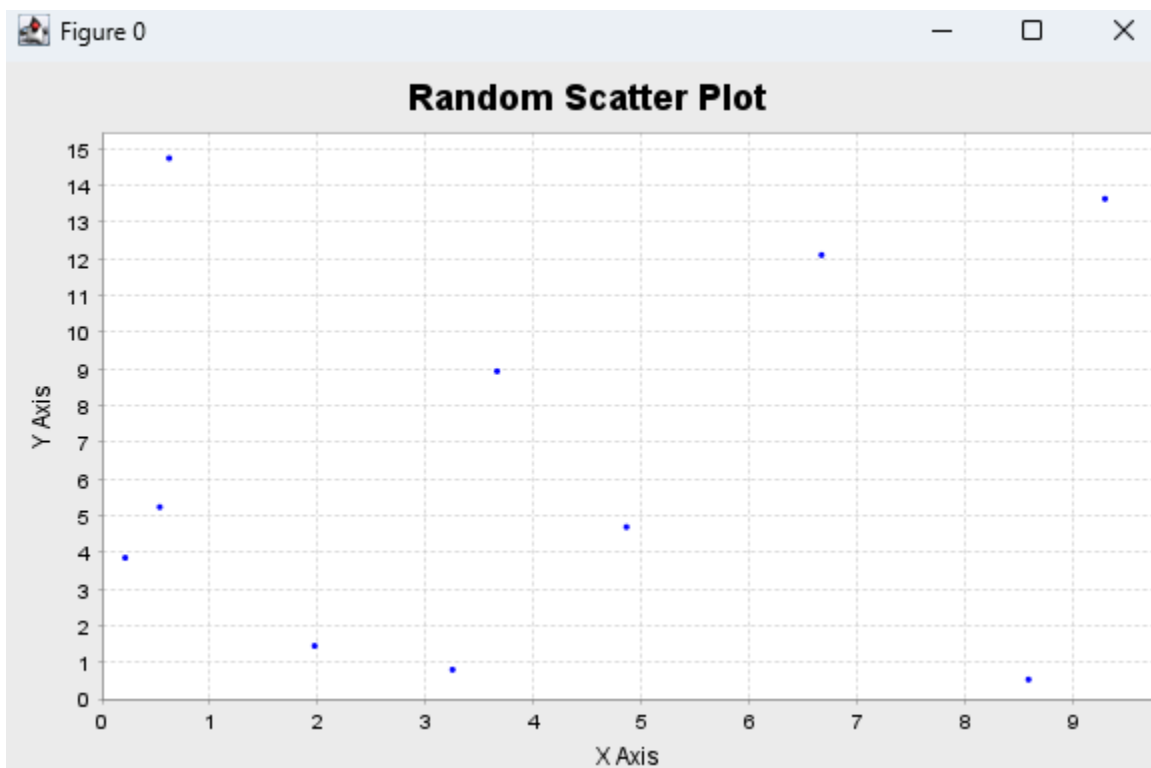
  p += plot(x, y, '.', colorcode = "blue")

  p.xlabel = "X Axis"
  p.ylabel = "Y Axis"
  p.title = "Random Scatter Plot"

  f.refresh()
  println("S007 DHRUV")
}

```

OUTPUT:



Practical no.17

17) Generate a histogram of a dataset using Breeze-viz. Experiment with different bin sizes.

INPUT:

```
import breeze.plot._
import scala.io.Source

object PRAC14 extends App {
  val bufferedSource = Source.fromResource("iris.csv")
  val data = bufferedSource.getLines().drop(1).map { line =>
    val cols = line.split(",").map(_.trim)
    cols(0).toDouble
  }.toList
  bufferedSource.close()

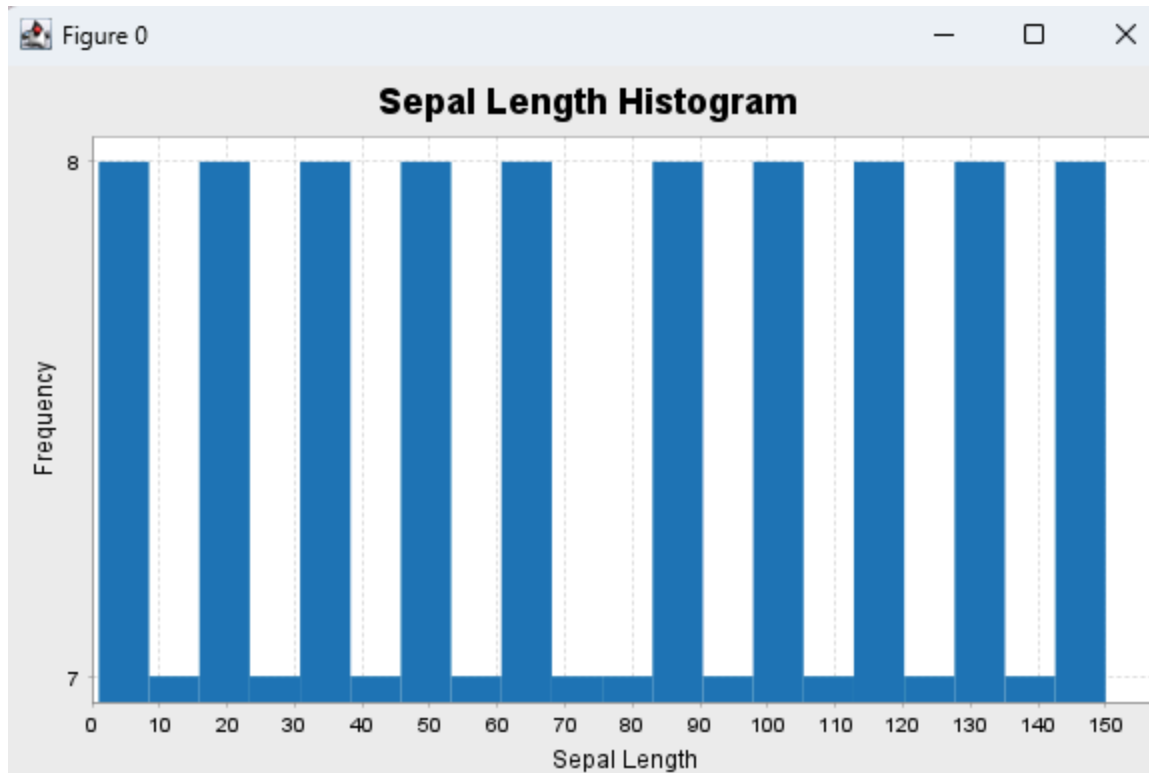
  val f = Figure()
  val p = f.subplot(0)

  p += hist(data, bins = 20)

  p.xlabel = "Sepal Length"
  p.ylabel = "Frequency"
  p.title = "Sepal Length Histogram"

  f.refresh()
  println("S007 DHRUV")
}
```

OUTPUT:



Practical no.18

18) Plot a line graph for a dataset showing a trend over time.

INPUT:

```

import breeze.plot._
import scala.io.Source

object PRAC14 extends App {
  val file = Source.fromResource("iris.csv")

  val x = file.getLines()
    .drop(1)
    .map(line => line.split(",")(0).toDouble)
    .toList

  file.close()

  val y = x.indices.map(_._2.toDouble).toList

  val fig = Figure()
  val p = fig.subplot(0)

  p += plot(y, x)

  p.xlabel = "Record Number"
  p.ylabel = "Sepal Length"
  p.title = "Sepal Length Trend"

  fig.refresh()
  println("S007 DHRUV")
}

```

OUTPUT:

